Desalination LLM: Large Language Model Agents Design Efficient Graphene Membranes for Water Desalination

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Abstract

Two-dimensional nanomaterials, particularly graphene, have emerged as promising candidates for next-generation desalination membranes because of their atomic thickness and exceptional transport properties. Yet the design of nanopores that simultaneously maximize water permeability and maintain high ion rejection remains a major challenge, as conventional optimization typically requires large numbers of costly simulations or experiments. Here, we present an autonomous large language model (LLM)-driven design framework for the iterative optimization of graphene nanopores. In this approach, LLM agents propose pore-shaping actions, evaluate candidate geometries through specialized computational tools, and refine designs in a closed-loop workflow. We benchmarked two leading model families, Google DeepMind Gemini and OpenAI GPT, using a structured chain-of-thought strategy combined with custom function toolkits for atomistic pore modification. At each design step, candidate carbon-atom removals were assessed using pre-trained deep learning predictors of water

flux and ion rejection, while manufacturability constraints, including the avoidance of unstable dangling bonds, were explicitly enforced. Our results show that coupling LLM reasoning with domain-specific tools enables precise and reliable control over nanopore evolution and accelerates the identification of high-performance membrane configurations. This work establishes LLM-guided autonomous design as a promising route for the discovery and optimization of advanced desalination membranes.

Keywords

Machine learning, agent, 2D materials, LLM, Gemini, GPT, graphene, nanopore, water flux, ion rejection

Introduction

Water scarcity is one of today's most urgent global challenges, driven by population growth, climate change, and dwindling freshwater reserves.[1] Desalination offers a vital path to tackle the world's vast seawater supply, yet existing technologies struggle to balance high water production, salt rejection, and energy efficiency.[2–6] Graphene-based membranes have emerged as a frontrunner in next-generation desalination technologies due to their exceptional strength, atomic thinness, and tunable nanopore properties.[7,8] Graphene nanopores offer a promising route for water desalination, but designing nanopore geometry remains a difficult optimization task - high water flux must be achieved without compromising ion rejection.[9] Fabricating 2D membranes, such as graphene, for every design trial is both costly and time-consuming, as each pore requires complex nanofabrication and meticulous laboratory testing to measure water flux and salt rejection. Running an experimental process across hundreds of candidate pores would take months and enormous expensive resources. Instead, molecular dynamics (MD) simulations offer a practical way to test designs computationally, capturing atomistic details of water and ion transport without the expense of physical fabrication.[10–13] Tradi-

tional molecular dynamics (MD) simulations provide precise water and salt transport data, but they are computationally expensive and resource-intensive. For example, a 10 ns run on a 4 nm $\times$ 4 nm graphene pore requires approximately 36 hours on a 56-core CPU cluster, meaning that screening geometrically optimized membranes among 3,000 pores would take over 12 years.[14]

To address this bottleneck, surrogate models, which can quickly predict water flux and ion rejection, have been developed, enabling rapid nanopore screening without the heavy cost of full MD simulations.[14] Yet, CNN and deep reinforcement learning (DRL) alone cannot create and interpret new pore shapes wisely. This brings us to a key question: since the number of possible atom removals is enormous, "can we find a smarter way to selectively remove atoms and get the best and fastest optimized design pore geometries as an agent?"

Addressing this challenge points us toward the intersection of artificial intelligence and materials science, where LLMs are no longer limited to passive text generation but are emerging as autonomous agents capable of driving real-world engineering tasks.[11,14–16]

To harness the full potential of LLMs in specialized fields, researchers have begun exploring domain-specific training and fine-tuning strategies—efforts that have already yielded promising results. For instance, in the field of engineering, LLM-Based Multi-Agent Systems for Software Engineering have explored the transformative potential of integrating Large Language Models into Multi-Agent (LMA) systems for addressing complex challenges in software engineering (SE).[17] Large Language Model Agent as a Mechanical Designer is a powerful new class of reasoning-driven, language-guided optimizers capable of autonomous design and iterative structural refinement.[18] MD simulations for Proteins using LLM can reduce setup time, minimize manual errors, and recommend a scalable solution for handling multiple protein systems in parallel.[19] Retrieval-Augmented Generation (RAG), which integrates data from diverse sources, offers a powerful tool for streamlining and enhancing literature retrieval, accelerating research in nanotechnology using LLM.[20]

Recent advances in LLM architectures, such as OpenAI's GPT-5[21]and Google Deep-

Mind's Gemini[22] models, have opened the door to a new class of intelligent agents capable of sophisticated reasoning, planning, and decision-making. Large Language Model agents are evolving beyond static tool use by directly executing dynamic, interpretable Python code. This shift, exemplified by CodeAct, unlocks a unified and flexible action space that empowers agents to reason, revise, and autonomously solve complex real-world tasks.[23] These models, when paired with tools like function calling and (CLI), can now go beyond language understanding to orchestrate entire workflows across scientific domains. The command-line accessibility of the Gemini model, in particular, has empowered developers to script fine-grained control pipelines, enabling LLMs to make context-aware decisions, execute simulation commands, and adapt their strategies based on real-time feedback.

In this work, we introduce a novel framework that autonomously designs nanopores as a sequential decision-making problem driven by Multiple LLM agents. Rather than manually iterating through atom configurations, we equip the Gemini model, accessed via its CLI, with a suite of custom functions that allow it to analyze, design, reason, and act within a defined molecular design space. Through a COT[24] prompting strategy, the model is guided to remove or preserve atoms in a graphene lattice, with each step evaluated by pre-trained neural networks that predict water flux and ion rejection metrics. Crucially, the model also adheres to manufacturability constraints, such as avoiding unstable dangling bonds, ensuring that the proposed designs are not only optimal in performance but also physically feasible. By benchmarking the performance of Gemini and ChatGPT, we highlight how AI can close the gap between abstract scientific reasoning and the practical execution of complex computational tasks. This shift paves the way for a new era of AI-assisted materials engineering, where advanced models accelerate desalination research and drive improved membrane performance.

Methodology

In this section, we present the methodology behind developing the Desalination LLM, with a particular focus on the overall pipeline framework and the step-by-step design process of the graphene nanopore.

LLM Desalination Framework

The objective of our system is to design graphene nanopores for water desalination, optimizing for high water flux and ion rejection through an iterative, DRL-inspired process that emphasized "Ozark-inspired" and "colonized" pore growth with strict atom removal constraints.[11] This study utilized pre-trained CNNs for graphene image feature extraction and Multi-Layer Perceptrons (MLPs) for predicting flux and rejection from these features.[14] The proposed "Desalination LLM" acts as an autonomous, intelligent designer for membrane optimization. As illustrated in Figure 1, the core language model processes complex structural and chemical data to direct the precise, nanoscale modification of filtration pores. The conceptual manipulator represents the LLM's execution of targeted atomic removals. By iteratively refining the membrane architecture based on predictive feedback, the AI framework bridges the gap between theoretical pore design and optimal physical performance, effectively separating ions to yield high-purity water.(Figure 1a).

Desalination Design Process

The design process is structured as an iterative loop, where the graphene membrane evolves over a series of "steps." Each step involves exploring multiple potential pore configurations ("trials") and selecting the most promising one based on predicted water flux and ion rejection performance metrics. The design process begin by input use query prompt to add series of instructions to the LLM agents.(Figure 2):

Initial State: The simulation begins with a pristine graphene sheet, represented by a Protein

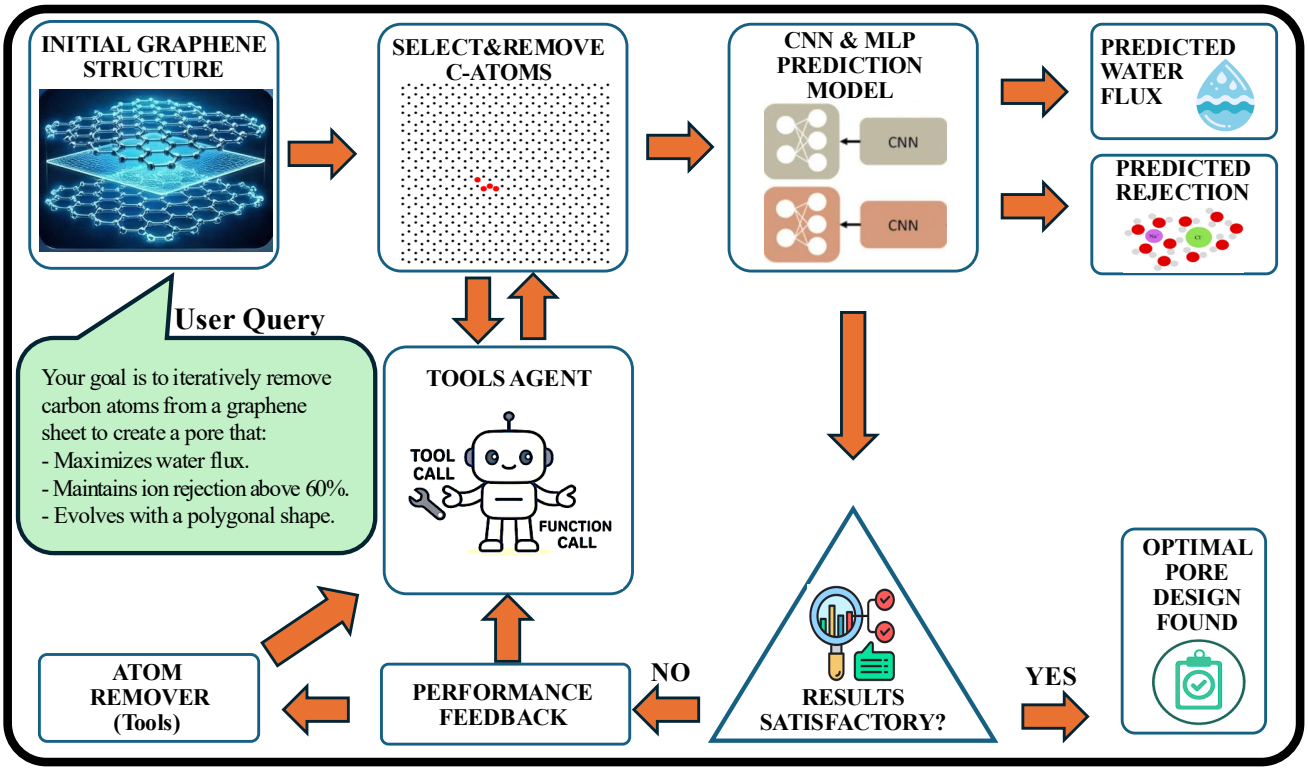


Figure 1: The overall framework of the LLM-guided graphene 2d nanomembrane water desalination design. Schematics showing the general workflow: user query prompt, the pretrained CNN model (Deep reinforcement learning framework proposed by Wang et al.[14]), The schematic depicts an artificial intelligence agent (central neural network) autonomously directing the precision engineering of a nanoporous filtration membrane. Through this guided architectural refinement, the system optimizes the separation of the ions to achieve highly efficient, high-purity water permeation.

Data Bank (PDB)[25] file. At this initial state, no atoms have been removed, resulting in zero water flux and 100% ion rejection a theoretical baseline for an impermeable membrane.[26] This serves as the starting point for pore evolution.

Atom Removal Strategy: The core of the design process involves strategically removing carbon atoms to form nanopores. Unlike arbitrary removal, this methodology employs a controlled, "Ozark-inspired" approach[11] following process below:

*Iterative Removal from the Center: Pore formation initiates from the center of the graphene sheet. This ensures that the pore grows organically and maintains a contiguous structure, avoiding isolated atom removals that might lead to unstable or non-functional pores.

Closest Neighborhood Selection (k parameter): To promote a "colonized" and irregular pore

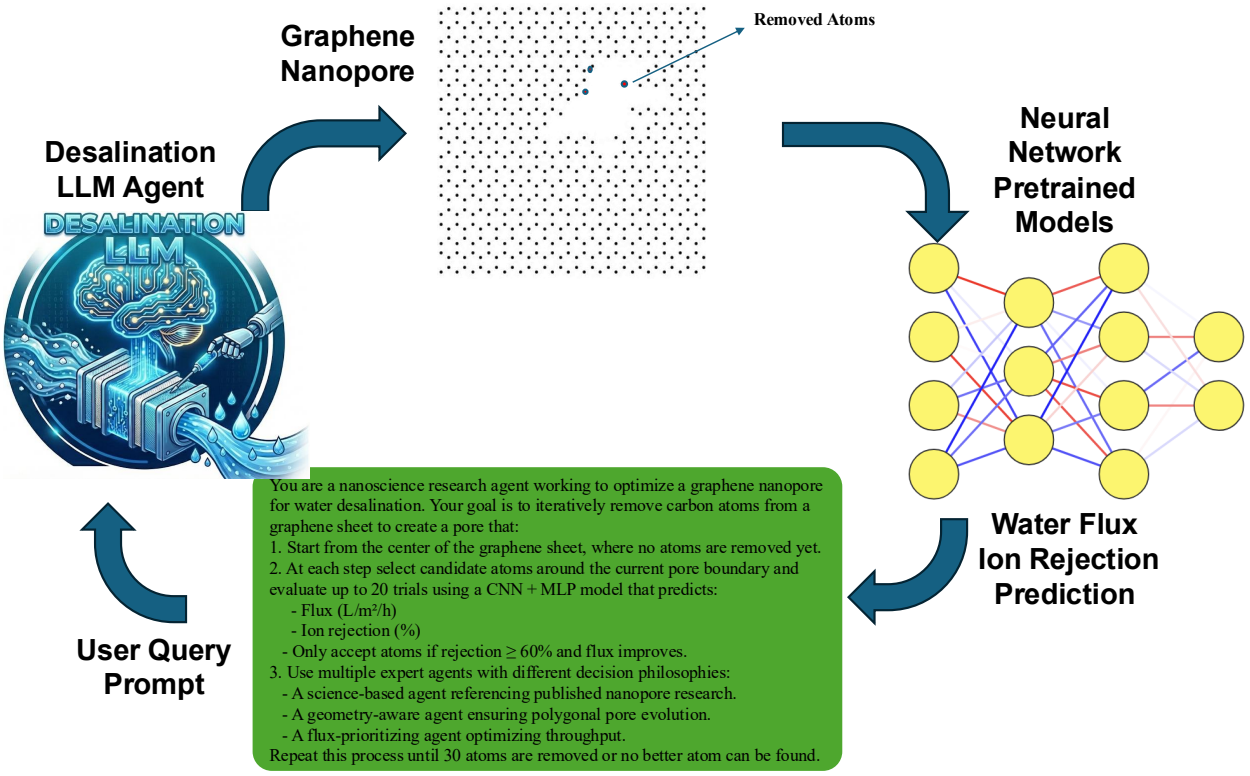


Figure 2: Process flow of the proposed Desalination LLM framework. The system utilizes prompt-driven instructions to guide specialized LLM agents through an iterative design loop, ultimately predicting and optimizing water flux and ion rejection rates.

shape, candidate atoms for removal are selected from the nearest neighborhood of already removed atoms. To achieve better accuracy and determine which k parameter is more effective in terms of water flux and ion rejection, we set three k parameters (k=15, k=30, and k=45). This approach mimics a more realistic etching or defect propagation process, where new removals are likely to occur adjacent to existing ones. This is crucial for manufacturability, as it minimizes "dangling bonds" and promotes chemically stable pore edges.[14,27,28]

*Trial-Based Exploration: At each design step, multiple trials are conducted (Trials=10, Trials=20, and Trials=30). In each trial, a small number of atoms are randomly selected from the candidate pool and hypothetically removed. This allows the agent to explore a diverse set of pore configurations at each iteration, rather than committing to a single removal path. The inclusion of "0 atoms removed" in a trial provides flexibility, allowing the agent to maintain the current pore status if no better configuration is found by removing additional atoms.

* Strict Atom Addition Constraint: A critical refinement in the design process is the strict

enforcement of maximum removed atoms (2, 3, or 5 atoms). This parameter limits the total number of new atoms added to the pore in a single step in each trial, including those added during the "colonization" process. This constraint is vital for controlling the growth rate and ensuring that the pore evolution remains gradual and manageable.

* "Colonized" and Irregular Pore Shapes: The design explicitly aims for irregular, polygon-like pore shapes, often referred to as "Ozark-inspired".[11] This is achieved by identifying and removing any unremoved atoms that fall within the convex hull of the already removed atoms. This "colonization" ensures that the pore remains continuous and free of isolated graphene islands within its boundary, which is essential for both manufacturability and efficient water transport. The shapes can lead to lower energy barriers for water permeation and unique structural properties beneficial for desalination.[10,29]

Performance Evaluation: The performance of each generated nanopore is evaluated using pre-trained machine learning models as illustrated below (Figure 2):[14]

*Flux and Rejection Prediction: Two separate Multi-Layer Perceptron (MLP) models, trained on a dataset of graphene nanopores, predict the water flux and ion rejection rate. These models take an image representation of the nanopore as input, allowing for rapid evaluation without computationally expensive MD simulations.

* Trade-off and Selection Criteria: The primary objective is to achieve high water flux while maintaining a high ion rejection rate. A minimum rejection threshold is set to ensure practical desalination performance. At each step, the agent selects the trial that offers the highest water flux, provided its ion rejection rate meets or exceeds the minimum rejection threshold. If no trial meets this criterion, the agent prioritizes trials that improve the rejection rate. This ensures a balanced optimization strategy.

Simulation Parameters: The overall simulation design process is governed by the number of steps (The best achieved number of steps for this study is 30 steps), and maximum removals (The maximum total number of atoms that can be removed from the graphene sheet throughout the entire design process. This acts as an upper limit on pore size).

Results and Discussion

This study employed an iterative, AI-driven approach to engineer graphene nanopores for water desalination, aiming to maximize water flux while maintaining high ion rejection rates. The design process simulated the sequential removal of carbon atoms from a pristine graphene sheet, with each "step" representing a refinement of the pore structure (Each step does 20 trials among the 30 nearest neighbors of the previously removed atoms iteratively). We investigate two LLM models. The first one is GPT-5 from the OpenAI company,[30] and the second one is related to Gemini CLI from the Google DeepMind company.[22] GPT-5 is a text-based language model. It does not have direct access to the local file system or the ability to execute code on the machine. Its interaction model is purely conversational.[30] It generates text, including code snippets, explanations, and instructions, which, by adding several prompts, can make a functional code that interprets the water flux and rejection rate. On the other hand, Gemini CLI (Using Gemini 2.5 pro) operates as an interactive agent directly within a command-line environment. This means it has direct access to the file system and can execute shell commands and python scripts on machine.[22] To achieve better accuracy and assess the performance of the two models in terms of water desalination efficiency, we compare the water flux and ion rejection of the models at each step. Each step allows the removal of a maximum of one atom per trial from the graphene nanopore sheet. The main prompt given to the two models is the same. As the results show, for a certain amount of atom removal, the water flux in the Gemini model lines a smoother and incremental path, especially in the range of 150–170 ns^{-1}, with each step increase. At the same time, it can maintain more ion rejection around 95 %, otherwise in GPT-5 we will see the same water flux while the ion rejection rate falls below ≈95% (Figure3).

To ensure our results were both reliable and broadly representative, we repeated the Gemini-driven atom-removal protocol five times, using the identical prompt and chain-of-thought instructions. At every design step, we carried out twenty independent trials, each selecting up to three atoms from the thirty nearest neighbors of the last-removed site. By averaging the outcomes across these runs, we effectively mitigate the randomness inherent in trial-by-trial atom selection, ensuring that no single lucky removal of atoms skews our overall conclusions (Figure4a). To illustrate

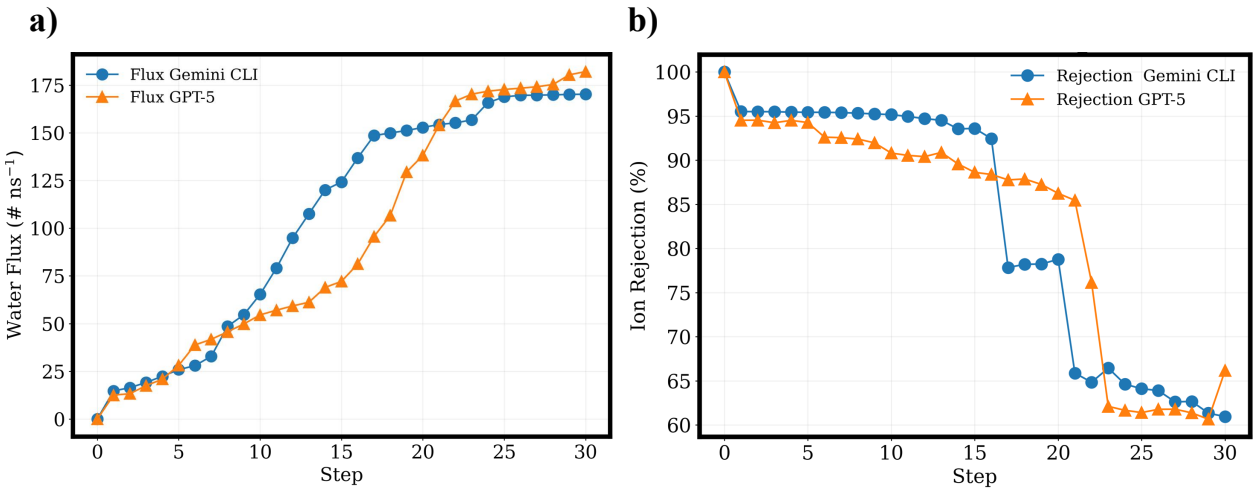


Figure 3: Gemini CLI and GPT-5 LLM models comparison using the same variables (trials per step each step=20, number of k-neighbours=30, maximum number of allowed removed atoms=1). **(a)**: Water flux vs number of steps. **(b)**: Ion rejection rate vs number of steps.

our design principles, we include schematic cartoons of the two top-performing LLM-generated nanopore geometries—both achieve high water flux and strong ion rejection (see Supporting Information Figure S1). In this diagram, ions (potassium in yellow, chloride in orange) carry hydration shells made of water molecules (oxygen in gray, hydrogen in green). The combined ion–shell are too large to squeeze through the narrow cavities of the pore; the red "X" regions mark the spots where the nanopore edges sterically block ion passage. The water flux histogram illustrates that, at a baseline ion rejection threshold of 90%, the LLM agent framework can extrapolate beyond the training dataset and DRL framework,[14] enabling the discovery of graphene nanopores with consistently higher water flux while maintaining ion rejection more than 90% than those identified by either the baseline dataset model (Figure4b). From the mentioned multi-run dataset, we calculate the mean and standard deviation of key metrics (water flux, ion rejection, and total atoms removed) at each stage (see Supporting Information Figure S2).

The machine learning models (MLPs) in this study extract 64-dimensional feature vectors from the nanopore images, which it impossible to comprehend patterns in 64 dimensions directly. t-SNE[31] effectively compresses this complex information into a 2D plot, allowing us to visually identify clusters, trends, and relationships among the nanopore designs based on their underlying

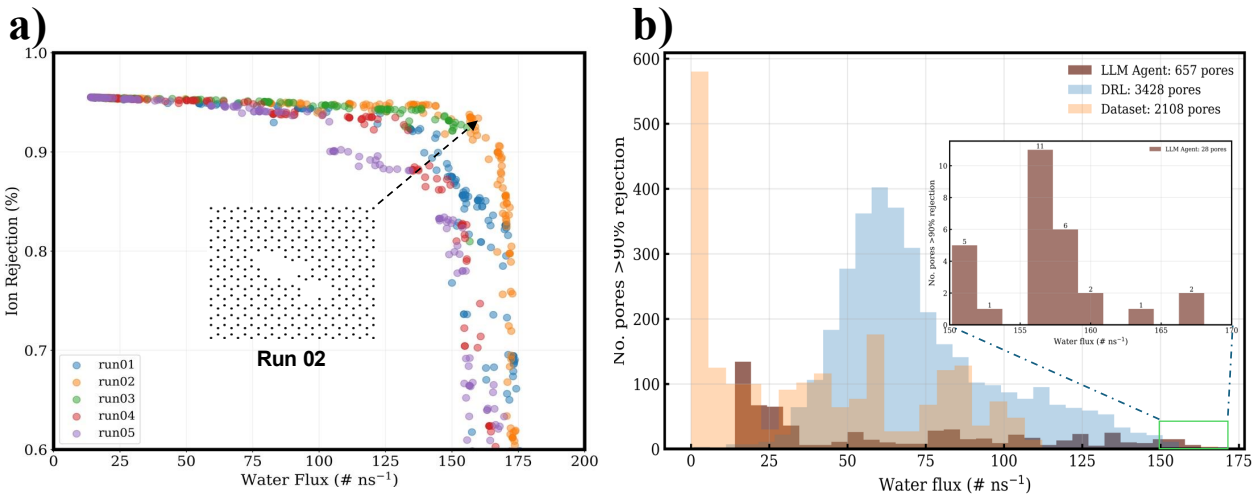


Figure 4: Geometry optimization characteristics of desalination performance: **(a)** Performance window for graphene nanopore desalination. The window 150–170 ns^{-1} achieves high water flux while preserving ion rejection at $\approx 95\%$, ideal for applications demanding strict purity (Run 02). **(b)** Histogram showing the distribution of water flux for nanopores with ion rejection rates above 90%, comparing results across three different models. The inset highlights LLM-agent–designed nanopores in the range 150–170 ns^{-1}, illustrating the details of the extrapolation-driven optimization process.

features (Figure 5). The color-coding of the t-SNE points by water flux and ion rejection (containing a 3000-generated dataset) allows us to directly observe how structural features (represented by their position in the t-SNE plot) correlate with functional performance. We can identify regions in the embedding space that correspond to high flux, high rejection (depicted by light yellow), or the desired trade-off, providing intuitive insights into the structure-property relationships. As shown in Figures 5a and b, all points are widely clustered; the spread-out distribution could indicate a broader range of designs. It can also hint at novel designs that occupy previously unexplored regions of the feature space.

Conclusion

This iterative, AI-driven graphene nanopore design agent provides a robust and scientifically grounded methodology for engineering high-performance desalination membranes. By combining a controlled, "Ozark-inspired" atom removal strategy with machine learning-based performance prediction and comprehensive multi-run statistical analysis, the agent can efficiently explore the

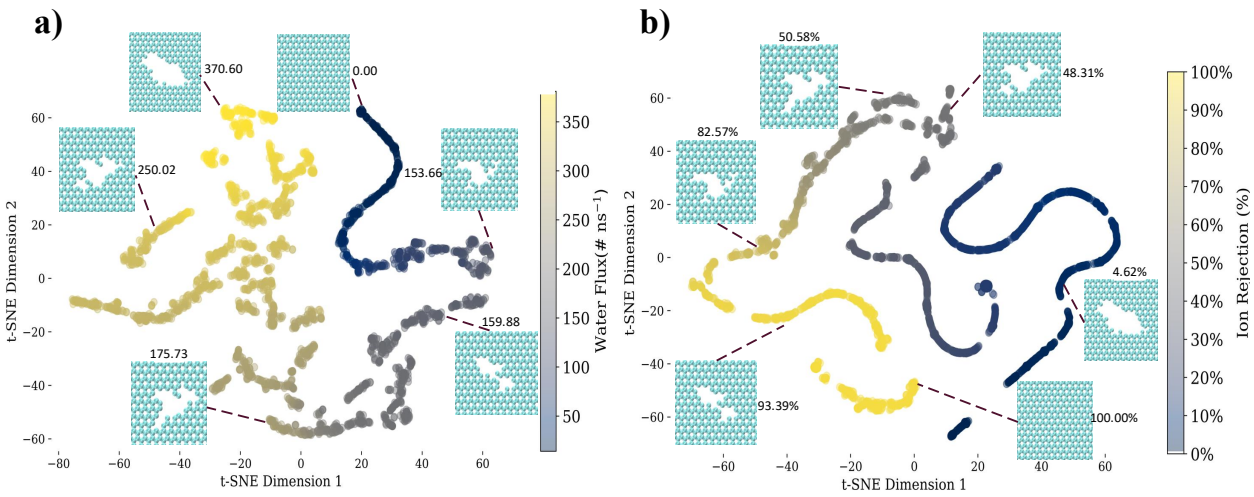


Figure 5: Feature Embedding from LLMs: By plotting t-SNE, we can visually assess how effectively our iterative design process explores the vast design space: **(a)** t-SNE visualization of LLM-derived feature embeddings for water flux prediction. Each point corresponds to a sample in the 2D embedding space and is colored by its predicted water flux, revealing distinct clusters of similar flux behavior. **(b)** t-SNE visualization of LLM-derived feature embeddings for ion rejection prediction. Each point represents a sample in the 2D embedding space and is color-coded by its predicted ion rejection, highlighting clusters of similar selectivity behavior.

vast design space of graphene nanopores.

By prioritizing manufacturable pore shapes, maintaining strict control over pore growth, and conducting detailed performance evaluations supported by statistical analysis, this work ensures that the generated nanopore designs are both scientifically robust and practically viable. Our LLM-agent–driven framework effectively identifies optimal trade-offs between water flux and ion rejection, offering a powerful approach to accelerate the discovery and development of advanced materials for water purification. Using this methodology, we extrapolated 28 previously unexplored nanopore designs beyond the training dataset, each achieving over 90% ion rejection while delivering water flux in the range of 150–170 ns^{-1}. This superior control enabled the Gemini-led agent to engineer a unique, non-circular pore geometry systematically. The final design is predicted to achieve an exceptional water flux of approximately 159.89 ns^{-1} while maintaining a high ion rejection rate of 93.4%. Continuous refinement of the design parameters—such as number of trials per step, maximum allowed atom removals per trial, and k-neighbor settings—enables fine-tuning of the exploration strategy to meet specific performance targets. Collectively, these advancements pave

the way for more efficient, high-performance, and sustainable solutions to global water desalination challenges.

Acknowledgement

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